Ecology

Diversity is not only a characteristic of living organisms but also of the content in biology textbooks, and ecology is the thread that gives us a holistic perspective to biology.

How biology is presented

- biology is presented either as botany, zoology and microbiology, or as classical and modern, the latter being a euphemism for molecular aspects of biology
- luckily we have many threads which weave the different areas of biological information into a unifying principle, and ecology is one such thread

What ecology explains

- the essence of biological understanding is to know how organisms, while remaining an individual, interact with other organisms and physical habitats as a group, and hence behave like organised wholes, that is population, community, ecosystem or even the whole biosphere
- a particular aspect of this is the study of anthropogenic environmental degradation and the socio-political issues it has raised
- this unit describes as well as takes a critical view of the above aspects

In this unit

Ch 11 Organisms and Populations

Ch 12 Ecosystem

Ch 13 Biodiversity and Conservation

Ramdeo Misra, 1908 to 1998

- he is revered as the Father of Ecology in India, born on 26 August 1908
- he obtained a Ph.D in Ecology in 1937 under Prof. W. H. Pearsall, FRS, from Leeds University in the UK
- he established teaching and research in ecology at the Department of Botany of the Banaras Hindu University, Varanasi

The research and the teaching

- his research laid the foundations for understanding of tropical communities and their succession, environmental responses of plant populations, and productivity and nutrient cycling in tropical forest and grassland ecosystems
- Misra formulated the first postgraduate course in ecology in India
- over 50 scholars obtained a Ph. D degree under his supervision and moved on to other universities and research institutes to initiate ecology teaching and research across the country

The honours, and what followed

- he was honoured with the Fellowships of the Indian National Science Academy and the World Academy of Arts and Science, and the prestigious Sanjay Gandhi Award in Environment and Ecology
- due to his efforts the Government of India established the National Committee for Environmental Planning and Coordination in 1972
- that committee, in later years, paved the way for the establishment of the Ministry of Environment and Forests in 1984

11.1 How and why

The levels of organisation

- our living world is fascinatingly diverse and amazingly complex, and we can try to understand its complexity by investigating processes at various levels of biological organisation
- those levels are macromolecules, cells, tissues, organs, individual organisms, populations, communities, ecosystems and biomes

The two types of question

- at any level of biological organisation we can ask two types of question, so when we hear the bulbul singing early morning in the garden we may ask how the bird sings, or why the bird sings
- the how-type questions seek the mechanism behind the process, while the why-type questions seek the significance of the process
- for the first question the answer might be in terms of the operation of the voice box and the vibrating bone in the bird, whereas for the second the answer may lie in the bird's need to communicate with its mate during breeding season

More questions of both types

- why night-blooming flowers are generally white
- how the bee knows which flower has nectar
- why cactus has so many thorns
- how the chick spures recognise her own mother

11.2 Populations

What a population is

- ecology is a subject which studies the interactions among organisms, and between the organism and its physical, that is abiotic, environment, and it is basically concerned with four levels of biological organisation, organisms, populations, communities and biomes
- in nature we rarely find isolated single individuals of any species, the majority of them living in groups in a well defined geographical area, sharing or competing for similar resources, potentially interbreeding and thus constituting a population, while a group of individuals resulting from asexual reproduction is also generally considered a population for the purpose of ecological studies
- all the cormorants in a wetland, rats in an abandoned dwelling, teakwood trees in a forest tract, bacteria in a culture plate and lotus plants in a pond are some examples
- although an individual organism is the one that has to cope with a changed environment, it is at the population level that natural selection operates to evolve the desired traits, so population ecology is an important area because it links ecology to population genetics and evolution

11.3 Population attributes

Birth rates and death rates

- a population has certain attributes whereas an individual organism does not, so an individual may have births and deaths but a population has birth rates and death rates, which refer to per capita births and deaths, the rates being expressed as change in numbers with respect to members of the population
- if in a pond there were 20 lotus plants last year and through reproduction 8 new plants are added, taking the current population to 28, we calculate the birth rate as 8/20 = 0.4 offspring per lotus per year
- if 4 individuals in a laboratory population of 40 fruitflies died during a specified time interval, say a week, the death rate during that period is 4/40 = 0.1 individuals per fruitfly per week

Sex ratio and age pyramid

- another attribute characteristic of a population is sex ratio, since an individual is either a male or a female but a population has a sex ratio, for example 60 per cent of the population female and 40 per cent male
- a population at any given time is composed of individuals of different ages, and if the age distribution is plotted for the population the resulting structure is called an age pyramid
- for human population the age pyramids generally show age distribution of males and females in a diagram, and the shape of the pyramids reflects the growth status of the population, whether it is growing, stable or declining

A growing population, broad at the base

A stable population, with straighter sides

A declining population, narrow at the base

Males and females shown on the two sides

Figure 11.A Representation of age pyramids for human population

11.4 Population density

How size is measured

- the size of the population tells us a lot about its status in the habitat, and whatever ecological processes we wish to investigate, be it the outcome of competition with another species, the impact of a predator or the effect of a pesticide application, we always evaluate them in terms of any change in the population size
- the size in nature could be as low as <10, as with Siberian cranes at Bharatpur wetlands in any year, or go into millions, as with Chlamydomonas in a pond
- population size, technically called population density and designated as N, need not necessarily be measured in numbers only, and although total number is generally the most appropriate measure it is in some cases either meaningless or difficult to determine

When numbers mislead

- in an area, if there are 200 carrot grass plants but only a single huge banyan tree with a large canopy, stating that the population density of banyan is low relative to that of carrot grass amounts to underestimating the enormous role of the banyan in that community, so in such cases the per cent cover or biomass is a more meaningful measure
- total number is again not an easily adoptable measure if the population is huge and counting is impossible or very time-consuming
- sometimes, for certain ecological investigations, there is no need to know the absolute population densities, relative densities serving the purpose equally well, so the number of fish caught per trap is a good enough measure of total population density in the lake, and we are mostly obliged to estimate population sizes indirectly, the tiger census in our national parks and tiger reserves often being based on pug marks and fecal pellets

Natality and immigration, adding to the density

Mortality and emigration, taking away from it

The four processes acting on the population at once

Figure 11.B

11.5 Population growth

The four processes

- the size of a population for any species is not a static parameter, it keeps changing with time depending on various factors including food availability, predation pressure and adverse weather, and it is these changes in population density that give us some idea of what is happening to the population
- natality refers to the number of births during a given period in the population that are added to the initial density, and mortality is the number of deaths during a given period
- immigration is the number of individuals of the same species that have come into the habitat from elsewhere during the time period under consideration, and emigration is the number of individuals of the population who left the habitat and gone elsewhere during that period

The equation

- if N is the population density at time t, then its density at time t +1 is $N_{t+1} = N_t + [(B + I) – (D + E)]$
- population density will increase if the number of births plus the number of immigrants is more than the number of deaths plus the number of emigrants
- under normal conditions births and deaths are the most important factors influencing population density, the other two assuming importance only under special conditions, so if a new habitat is just being colonised immigration may contribute more significantly to population growth than birth rates

Curve a, exponential, when resources do not limit growth

A J-shaped rise without limit

Curve b, logistic, when resources do limit growth

A sigmoid rise flattening at K, the carrying capacity

Figure 11.C Population growth curve a when responses are not limiting the growth, plot is exponential, b when responses are limiting the growth, plot is logistic, K is carrying capacity

11.6 Growth models

Exponential growth

- resource availability, that is food and space, is obviously essential for the unimpeded growth of a population, and ideally, when resources in the habitat are unlimited, each species has the ability to realise fully its innate potential to grow in number, as Darwin observed while developing his theory of natural selection, so the population grows in an exponential or geometric fashion
- if in a population of size N the per capita birth rates are represented as b and the per capita death rates as d, then the increase or decrease in N during a unit time period t is $dN/dt = (b – d) \times N$, and letting $(b–d) = r$, $dN/dt = rN$
- the r in this equation is called the intrinsic rate of natural increase and is a very important parameter chosen for assessing impacts of any biotic or abiotic factor on population growth
- for the Norway rat the r is 0.015 and for the flour beetle it is 0.12, while in 1981 the r value for human population in India was 0.0205

The integral form

- the equation describes the exponential or geometric growth pattern of a population and results in a J-shaped curve when we plot N in relation to time
- the integral form of the exponential growth equation is $Nt = N0$ ert, where Nt is population density after time t, N0 is population density at time zero, r is the intrinsic rate of natural increase and e is the base of natural logarithms, 2.71828
- any species growing exponentially under unlimited resource conditions can reach enormous population densities in a short time

11.7 How fast exponential growth is

The king and the minister

- Darwin showed how even a slow growing animal such as the elephant could reach enormous numbers in the absence of checks
- the king and the minister sat for a chess game, and the king, confident of winning, was ready to accept any bet, so the minister humbly said that if he won he wanted only some wheat grains, the quantity to be calculated by placing on the chess board one grain in square 1, then two in square 2, four in square 3, eight in square 4, and so on, doubling each time the previous quantity until all the 64 squares were filled
- the king accepted the seemingly silly bet and started the game, but unluckily for him the minister won, and the king, feeling that fulfilling the bet was so easy, started with a single grain on the first square and proceeded, but by the time he covered half the chess board he realised to his dismay that all the wheat produced in his entire kingdom pooled together would still be inadequate to cover all the 64 squares
- now think of a tiny Paramecium starting with just one individual and, through binary fission, doubling in numbers every day, and imagine what a mind-boggling population size it would reach in 64 days, provided food and space remain unlimited

11.8 Logistic growth

Carrying capacity

- no population of any species in nature has at its disposal unlimited resources to permit exponential growth, which leads to competition between individuals for limited resources, and eventually the fittest individual will survive and reproduce
- the governments of many countries have also realised this fact and introduced various restraints with a view to limit human population growth
- in nature a given habitat has enough resources to support a maximum possible number, beyond which no further growth is possible, and this limit is nature's carrying capacity (K) for that species in that habitat
- a population growing in a habitat with limited resources shows initially a lag phase, followed by phases of acceleration and deceleration and finally an asymptote, when the population density reaches the carrying capacity, and a plot of N in relation to time results in a sigmoid curve, this type of growth being called Verhulst-Pearl logistic growth

11.9 The logistic equation

The terms

- the equation is dN/dt = rN (K – N) divided by K, where N is population density at time t, r is the intrinsic rate of natural increase and K is the carrying capacity
- since resources for growth for most animal populations are finite and become limiting sooner or later, the logistic growth model is considered a more realistic one
- gathering from Government census data the population figures for India for the last 100 years, plotting them and checking which growth pattern is evident, is worth doing

11.10 Life history variation

Breeding strategies

- populations evolve to maximise their reproductive fitness, also called Darwinian fitness or a high r value, in the habitat in which they live, and under a particular set of selection pressures organisms evolve towards the most efficient reproductive strategy
- some organisms breed only once in their lifetime, such as Pacific salmon fish and bamboo, while others breed many times during their lifetime, such as most birds and mammals
- some produce a large number of small-sized offspring, such as oysters and pelagic fishes, while others produce a small number of large-sized offspring, such as birds and mammals
- ecologists suggest that life history traits of organisms have evolved in relation to the constraints imposed by the abiotic and biotic components of the habitat in which they live, and evolution of life history traits in different species is currently an important area of research

11.11 Population interactions

No species lives alone

- there is no natural habitat on earth inhabited just by a single species, and such a situation is even inconceivable, since for any species the minimal requirement is one more species on which it can feed
- even a plant species, which makes its own food, cannot survive alone, since it needs soil microbes to break down the organic matter in soil and return the inorganic nutrients for absorption, and it could not manage pollination without an animal agent
- so in nature animals, plants and microbes do not and cannot live in isolation but interact in various ways to form a biological community, and even in minimal communities many interactive linkages exist, although all may not be readily apparent
- interspecific interactions arise from the interaction of populations of two different species, and could be beneficial, detrimental or neutral to one of the species or both

11.12 The six interactions

Who gains and who loses

- both the species benefit in mutualism and both lose in competition in their interactions with each other
- in both parasitism and predation only one species benefits, the parasite and the predator respectively, and the interaction is detrimental to the other species, the host and the prey respectively
- the interaction where one species is benefitted and the other is neither benefitted nor harmed is called commensalism, while in amensalism one species is harmed whereas the other is unaffected
- predation, parasitism and commensalism share a common characteristic, that the interacting species live closely together

11.13 Predation

Energy up the trophic levels

- predation can be thought of as nature's way of transferring to higher trophic levels the energy fixed by plants, which raises the question of what would happen to all the energy fixed by autotrophic organisms if the community had no animals to eat the plants
- when we think of predator and prey it is most probably the tiger and the deer that come to mind, but a sparrow eating any seed is no less a predator, and although animals eating plants are categorised separately as herbivores, they are in a broad ecological context not very different from predators
- besides acting as conduits for energy transfer across trophic levels, predators keep prey populations under control, since but for predators prey species could achieve very high population densities and cause ecosystem instability
- when certain exotic species are introduced into a geographical area they become invasive and start spreading fast, because the invaded land does not have its natural predators, and the prickly pear cactus introduced into Australia in the early 1920s caused havoc by spreading rapidly into millions of hectares of rangeland, finally being brought under control only after a cactus-feeding predator, a moth, from its natural habitat was introduced

Table 11.A Population Interactions

Species A	Species B	Name of Interaction
+	+	Mutualism
–	–	Competition
+	–	Predation
+	–	Parasitism
+	0	Commensalism
–	0	Amensalism

11.14 What predators do, and what prey do back

Predators and diversity

- biological control methods adopted in agricultural pest control are based on the ability of the predator to regulate prey population
- predators also help in maintaining species diversity in a community, by reducing the intensity of competition among competing prey species
- in the rocky intertidal communities of the American Pacific Coast the starfish Pisaster is an important predator, and in a field experiment, when all the starfish were removed from an enclosed intertidal area, more than 10 species of invertebrates became extinct within a year, because of interspecific competition

Prudent predators, defended prey

- if a predator is too efficient and overexploits its prey, then the prey might become extinct and following it the predator will also become extinct for lack of food, which is the reason why predators in nature are prudent
- prey species have evolved various defenses to lessen the impact of predation, some species of insects and frogs being cryptically-coloured, that is camouflaged, to avoid being detected easily by the predator, while some are poisonous and therefore avoided
- the Monarch butterfly is highly distasteful to its predator, a bird, because of a special chemical present in its body, and interestingly the butterfly acquires this chemical during its caterpillar stage by feeding on a poisonous weed

Plants against herbivores

- for plants, herbivores are the predators, and nearly 25 per cent of all insects are known to be phytophagous, that is feeding on plant sap and other parts of plants, the problem being particularly severe for plants because, unlike animals, they cannot run away from their predators
- plants have therefore evolved an astonishing variety of morphological and chemical defences against herbivores, thorns, as in Acacia and cactus, being the most common morphological means of defence
- many plants produce and store chemicals that make the herbivore sick when they are eaten, inhibit feeding or digestion, disrupt its reproduction or even kill it, so the weed Calotropis, growing in abandoned fields, produces highly poisonous cardiac glycosides, which is why one never sees any cattle or goats browsing on this plant
- a wide variety of chemical substances that we extract from plants on a commercial scale, such as nicotine, caffeine, quinine, strychnine and opium, are produced by them actually as defences against grazers and browsers

11.15 Competition

What competition really is

- when Darwin spoke of the struggle for existence and survival of the fittest in nature he was convinced that interspecific competition is a potent force in organic evolution, and it is generally believed that competition occurs when closely related species compete for the same resources that are limiting, but this is not entirely true
- firstly, totally unrelated species could also compete for the same resource, as in some shallow South American lakes where visiting flamingoes and resident fishes compete for their common food, the zooplankton in the lake
- secondly, resources need not be limiting for competition to occur, since in interference competition the feeding efficiency of one species might be reduced due to the interfering and inhibitory presence of the other species, even if food and space are abundant
- therefore competition is best defined as a process in which the fitness of one species, measured in terms of its intrinsic rate of increase r, is significantly lower in the presence of another species

Evidence from nature

- it is relatively easy to demonstrate in laboratory experiments, as Gause and other experimental ecologists did, that when resources are limited the competitively superior species will eventually eliminate the other species, but evidence for such competitive exclusion occurring in nature is not always conclusive
- strong and persuasive circumstantial evidence does exist however in some cases, since the Abingdon tortoise in Galapagos Islands became extinct within a decade after goats were introduced on the island, apparently due to the greater browsing efficiency of the goats
- another evidence comes from what is called competitive release, where a species whose distribution is restricted to a small geographical area because of the presence of a competitively superior species is found to expand its distributional range dramatically when the competing species is experimentally removed
- Connell's elegant field experiments showed that on the rocky sea coasts of Scotland the larger and competitively superior barnacle Balanus dominates the intertidal area and excludes the smaller barnacle Chathamalus from that zone, while in general herbivores and plants appear to be more adversely affected by competition than carnivores

Gause's principle, and resource partitioning

- Gause's Competitive Exclusion Principle states that two closely related species competing for the same resources cannot co-exist indefinitely and the competitively inferior one will be eliminated eventually, which may be true if resources are limiting but not otherwise
- more recent studies do not support such gross generalisations about competition, and while they do not rule out the occurrence of interspecific competition in nature they point out that species facing competition might evolve mechanisms that promote co-existence rather than exclusion
- one such mechanism is resource partitioning, since if two species compete for the same resource they could avoid competition by choosing different times for feeding or different foraging patterns, and MacArthur showed that five closely related species of warblers living on the same tree were able to avoid competition and co-exist due to behavioural differences in their foraging activities

11.16 Parasitism

Host specificity and co-evolution

- considering that the parasitic mode of life ensures free lodging and meals, it is not surprising that parasitism has evolved in so many taxonomic groups from plants to higher vertebrates
- many parasites have evolved to be host-specific, that is able to parasitise only a single species of host, in such a way that both host and parasite tend to co-evolve, so that if the host evolves special mechanisms for rejecting or resisting the parasite, the parasite has to evolve mechanisms to counteract and neutralise them in order to be successful with the same host species
- in accordance with their life styles, parasites evolved special adaptations such as the loss of unnecessary sense organs, presence of adhesive organs or suckers to cling on to the host, loss of digestive system and high reproductive capacity
- the life cycles of parasites are often complex, involving one or two intermediate hosts or vectors to facilitate parasitisation of the primary host, so the human liver fluke, a trematode parasite, depends on two intermediate hosts, a snail and a fish, while the malarial parasite needs a vector, the mosquito, to spread to other hosts

The harm they do

- the majority of parasites harm the host, they may reduce the survival, growth and reproduction of the host and reduce its population density
- they might render the host more vulnerable to predation by making it physically weak
- whether an ideal parasite should be able to thrive within the host without harming it, and why natural selection did not lead to the evolution of such totally harmless parasites, are worth thinking through

Ecto and endoparasites

- parasites that feed on the external surface of the host organism are called ectoparasites, the most familiar examples being the lice on humans and ticks on dogs, while many marine fish are infested with ectoparasitic copepods
- Cuscuta, a parasitic plant commonly found growing on hedge plants, has lost its chlorophyll and leaves in the course of evolution and derives its nutrition from the host plant which it parasitises, while the female mosquito is not considered a parasite although it needs our blood for reproduction
- in contrast, endoparasites are those that live inside the host body at different sites, liver, kidney, lungs, red blood cells and so on, and their life cycles are more complex because of their extreme specialisation, their morphological and anatomical features being greatly simplified while emphasising their reproductive potential
- brood parasitism in birds is a fascinating example, in which the parasitic bird lays its eggs in the nest of its host and lets the host incubate them

A fig flower pollinated by the wasp

The wasp laying eggs inside the fig fruit

The fig giving some developing seeds as food for the larvae

A tight one-to-one relationship between the two

Figure 11.D Mutual relationship between fig tree and wasp: (a) Fig flower is pollinated by wasp; (b) Wasp laying eggs in a fig fruit

11.17 Brood parasitism and commensalism

The mimicked egg

- during the course of evolution the eggs of the parasitic bird have evolved to resemble the host's egg in size and colour, to reduce the chances of the host bird detecting the foreign eggs and ejecting them from the nest
- following the movements of the cuckoo, that is the koel, and the crow in a neighborhood park during the breeding season, from spring to summer, shows brood parasitism in action

Commensalism

- this is the interaction in which one species benefits and the other is neither harmed nor benefited, as with an orchid growing as an epiphyte on a mango branch, and barnacles growing on the back of a whale, which benefit while neither the mango tree nor the whale derives any apparent benefit
- the cattle egret and grazing cattle in close association is a classic example, since the egrets always forage close to where the cattle are grazing because the cattle, as they move, stir up and flush out insects from the vegetation that otherwise might be difficult for the egrets to find and catch
- another example is the interaction between the sea anemone, which has stinging tentacles, and the clown fish that lives among them, the fish getting protection from predators which stay away from the stinging tentacles while the anemone does not appear to derive any benefit by hosting the clown fish

A bee at an orchid flower

One petal resembling the female of the bee

The male bee dusted with pollen as it visits

Pollen transferred when it visits the next flower

Figure 11.E Showing bee-a pollinator on orchid flower

11.18 Mutualism

Lichens, mycorrhizae and figs

- this interaction confers benefits on both the interacting species, lichens representing an intimate mutualistic relationship between a fungus and photosynthesising algae or cyanobacteria, while the mycorrhizae are associations between fungi and the roots of higher plants, the fungi helping the plant in the absorption of essential nutrients from the soil while the plant in turn provides the fungi with energy-yielding carbohydrates
- the most spectacular and evolutionarily fascinating examples are found in plant-animal relationships, since plants need the help of animals for pollinating their flowers and dispersing their seeds, and animals have to be paid fees for those services, plants offering rewards in the form of pollen and nectar for pollinators and juicy and nutritious fruits for seed dispersers
- the mutually beneficial system should also be safeguarded against cheaters, for example animals that try to steal nectar without aiding in pollination, so plant-animal interactions often involve co-evolution of the mutualists, the evolution of the flower and its pollinator species being tightly linked with one another
- in many species of fig trees there is a tight one-to-one relationship with the pollinator species of wasp, meaning that a given fig species can be pollinated only by its partner wasp species and no other, the female wasp using the fruit not only as an oviposition site but using the developing seeds within the fruit for nourishing its larvae, pollinating the fig inflorescence while searching for suitable egg-laying sites, and in return for the favour of pollination the fig offers the wasp some of its developing seeds as food

Sexual deceit in an orchid

- orchids show a bewildering diversity of floral patterns, many of which have evolved to attract the right pollinator insect, bees and bumblebees, and ensure guaranteed pollination by it, though not all orchids offer rewards
- the Mediterranean orchid Ophrys employs sexual deceit to get pollination done by a species of bee, one petal of its flower bearing an uncanny resemblance to the female of the bee in size, colour and markings, so the male bee is attracted to what it perceives as a female, pseudocopulates with the flower, and during that process is dusted with pollen
- when this same bee pseudocopulates with another flower it transfers pollen to it and thus pollinates the flower, which shows how co-evolution operates, since if the female bee's colour patterns change even slightly for any reason during evolution, pollination success will be reduced unless the orchid flower co-evolves to maintain the resemblance of its petal to the female bee

11.19 Summary

Ecology and the population

- as a branch of biology, ecology is the study of the relationships of living organisms with the abiotic, that is physico-chemical, factors and the biotic components, that is other species, of their environment, and it is concerned with four levels of biological organisation, organisms, populations, communities and biomes
- evolutionary changes through natural selection take place at the population level and hence population ecology is an important area of ecology, a population being a group of individuals of a given species sharing or competing for similar resources in a defined geographical area
- populations have attributes that individual organisms do not, birth rates and death rates, sex ratio and age distribution, and the proportion of different age groups of males and females in a population is often presented graphically as an age pyramid, whose shape indicates whether a population is stationary, growing or declining
- ecological effects of any factors on a population are generally reflected in its size, that is population density, which may be expressed in different ways, numbers, biomass or per cent cover, depending on the species

Growth and interaction

- populations grow through births and immigration and decline through deaths and emigration, and when resources are unlimited the growth is usually exponential, but when resources become progressively limiting the growth pattern turns logistic, and in either case growth is ultimately limited by the carrying capacity of the environment, while the intrinsic rate of natural increase is a measure of the inherent potential of a population to grow
- in nature populations of different species in a habitat do not live in isolation but interact in many ways, and depending on the outcome these interactions between two species are classified as competition, where both species suffer, predation and parasitism, where one benefits and the other suffers, commensalism, where one benefits and the other is unaffected, amensalism, where one is harmed and the other unaffected, and mutualism, where both species benefit
- predation is a very important process through which trophic energy transfer is facilitated, and some predators help in controlling their prey populations, while plants have evolved diverse morphological and chemical defenses against herbivory
- in competition it is presumed that the superior competitor eliminates the inferior one, the Competitive Exclusion Principle, but many closely related species have evolved various mechanisms which facilitate their co-existence, and some of the most fascinating cases of mutualism in nature are seen in plant-pollinator interactions

11.20 Exercises

From this chapter

1. List the attributes that populations possess but not individuals.
2. If a population growing exponentially double in size in 3 years, what is the intrinsic rate of increase (r) of the population?
3. Name important defence mechanisms in plants against herbivory.
4. An orchid plant is growing on the branch of mango tree. How do you describe this interaction between the orchid and the mango tree?
5. What is the ecological principle behind the biological control method of managing with pest insects?
6. Define population and community.
7. Define the following terms and give one example for each: (a) Commensalism (b) Parasitism (c) Camouflage (d) Mutualism (e) Interspecific competition
8. With the help of suitable diagram describe the logistic population growth curve.
9. Select the statement which explains best parasitism. (a) One organism is benefited. (b) Both the organisms are benefited. (c) One organism is benefited, other is not affected. (d) One organism is benefited, other is affected.
10. List any three important characteristics of a population and explain.